

This supplementary document continues the appendix labeling of the main manuscript. Appendix C remains in the manuscript, while this supplement provides additional material associated with Appendices B and D.

B The detailed algorithm of nonlinear Kalman filters used in the paper

The detailed implementations of the proposed framework for EKF, EKF2, CKF, and UKF, together with the conventional IEKF benchmark used for comparison, are presented in Algorithms 1, 2, 3, 4, and 5, respectively. Replacing the “Recalibrate” and “Back out” steps in the proposed framework with (13) recovers the conventional framework.

Algorithm 1 The extended Kalman filter with the new framework

Input: Process noise covariance matrix $\mathbf{Q}_k$, Measurement noise covariance matrix $\mathbf{R}_k$, state transition function $\mathbf{f}(\mathbf{x}, \mathbf{u})$, measurement function $\mathbf{h}(\mathbf{x})$, system inputs $\mathbf{u}_k$, measurements $\mathbf{z}_k$

Initialization:

- 1: $\hat{\mathbf{x}}_{0|0} = \mathbb{E}[\mathbf{x}_0]$
- 2: $\mathbf{P}_{0|0} = \mathbb{E}[(\hat{\mathbf{x}}_{0|0} - \mathbf{x}_0)(\hat{\mathbf{x}}_{0|0} - \mathbf{x}_0)^T]$
- 3: **for** every time step k **do**
- Predict:**
- 4: $\mathbf{F}_k = \frac{\partial \mathbf{f}}{\partial \mathbf{x}}|_{\hat{\mathbf{x}}_{k-1|k-1}, \mathbf{u}_{k-1}}$
- 5: $\hat{\mathbf{x}}_{k|k-1} = \mathbf{f}(\hat{\mathbf{x}}_{k-1|k-1}, \mathbf{u}_{k-1})$
- 6: $\mathbf{P}_{k|k-1} = \mathbf{F}_k \mathbf{P}_{k-1|k-1} \mathbf{F}_k^T + \mathbf{Q}_k$
- Update:**
- 7: $\mathbf{H}_{k|k-1} = \frac{\partial \mathbf{h}}{\partial \mathbf{x}}|_{\hat{\mathbf{x}}_{k|k-1}}$
- 8: $\hat{\mathbf{z}}_{k|k-1} = \mathbf{h}(\hat{\mathbf{x}}_{k|k-1})$
- 9: $\mathbf{P}_{xz,k|k-1} = \mathbf{P}_{k|k-1} \mathbf{H}_{k|k-1}^T$
- 10: $\mathbf{P}_{z,k|k-1} = \mathbf{H}_{k|k-1} \mathbf{P}_{k|k-1} \mathbf{H}_{k|k-1}^T$
- 11: $\mathbf{S}_{k|k-1} = \mathbf{P}_{z,k|k-1} + \mathbf{R}_k$
- 12: $\mathbf{K}_k = \mathbf{P}_{xz,k|k-1} \mathbf{S}_{k|k-1}^{-1}$
- 13: $\hat{\mathbf{x}}_{k|k} = \hat{\mathbf{x}}_{k|k-1} + \mathbf{K}_k (\mathbf{z}_k - \hat{\mathbf{z}}_{k|k-1})$
- Recalibrate:**
- 14: $\mathbf{H}_{k|k} = \frac{\partial \mathbf{h}}{\partial \mathbf{x}}|_{\hat{\mathbf{x}}_{k|k}}$
- 15: $\mathbf{P}_{xz,k|k} = \mathbf{P}_{k|k-1} \mathbf{H}_{k|k}^T$
- 16: $\mathbf{P}_{z,k|k} = \mathbf{H}_{k|k} \mathbf{P}_{k|k-1} \mathbf{H}_{k|k}^T$
- 17: $\mathbf{S}_{k|k} = \mathbf{P}_{z,k|k} + \mathbf{R}_k$
- 18: $\mathbf{P}_{k|k} = \mathbf{P}_{k|k-1} + \mathbf{K}_k \mathbf{S}_{k|k} \mathbf{K}_k^T - \mathbf{P}_{xz,k|k} \mathbf{K}_k^T - \mathbf{K}_k \mathbf{P}_{xz,k|k}^T$
- Back out:**
- 19: **if** $\text{tr}(\mathbf{P}_{k|k}) > \text{tr}(\mathbf{P}_{k|k-1})$ **then**
- 20: $\hat{\mathbf{x}}_{k|k} = \hat{\mathbf{x}}_{k|k-1}$
- 21: $\mathbf{P}_{k|k} = \mathbf{P}_{k|k-1}$
- 22: **end if**
- 23: **end for**

Algorithm 2 The second-order extended Kalman filter with the new framework

Input: Process noise covariance matrix $\mathbf{Q}_k$, Measurement noise covariance matrix $\mathbf{R}_k$, states number n_x , measurements number n_m , state transition function $\mathbf{f}(\mathbf{x}, \mathbf{u})$, measurement function $\mathbf{h}(\mathbf{x})$, system inputs $\mathbf{u}_k$, measurements $\mathbf{z}_k$

Initialization:

- 1: $\hat{\mathbf{x}}_{0|0} = \mathbb{E}[\mathbf{x}_0]$
- 2: $\mathbf{P}_{0|0} = \mathbb{E}[(\hat{\mathbf{x}}_{0|0} - \mathbf{x}_0)(\hat{\mathbf{x}}_{0|0} - \mathbf{x}_0)^T]$
- 3: **for** every time step k **do**
- Predict:**
- 4: $\mathbf{F}_k = \frac{\partial \mathbf{f}}{\partial \mathbf{x}}|_{\hat{\mathbf{x}}_{k-1|k-1}, \mathbf{u}_{k-1}}$
- 5: $\mathbf{F}_{i,k}^* = [\frac{\partial^2 \mathbf{f}_i}{\partial x_m \partial x_n}|_{\hat{\mathbf{x}}_{k-1|k-1}, \mathbf{u}_{k-1}}]_{mn}, i = 1, \dots, n_x$
- 6: $\hat{\mathbf{x}}_{k|k-1} = \mathbf{f}(\hat{\mathbf{x}}_{k-1|k-1}, \mathbf{u}_{k-1}) + \frac{1}{2} [\text{tr}(\mathbf{F}_{i,k}^* \mathbf{P}_{k-1|k-1})]_i$
- 7: $\mathbf{P}_{k|k-1} = \mathbf{F}_k \mathbf{P}_{k-1|k-1} \mathbf{F}_k^T + \frac{1}{2} [\text{tr}(\mathbf{F}_{i,k}^* \mathbf{P}_{k-1|k-1} \mathbf{F}_{j,k}^* \mathbf{P}_{k-1|k-1})]_{ij} + \mathbf{Q}_k$
- Update:**
- 8: $\mathbf{H}_{k|k-1} = \frac{\partial \mathbf{h}}{\partial \mathbf{x}}|_{\hat{\mathbf{x}}_{k|k-1}}$
- 9: $\mathbf{H}_{i,k}^* = [\frac{\partial^2 \mathbf{h}_i}{\partial x_m \partial x_n}|_{\hat{\mathbf{x}}_{k|k-1}}]_{mn}, i = 1, \dots, n_m$
- 10: $\hat{\mathbf{z}}_{k|k-1} = \mathbf{h}(\hat{\mathbf{x}}_{k|k-1}) + \frac{1}{2} [\text{tr}(\mathbf{H}_{i,k}^* \mathbf{P}_{k|k-1})]_i$
- 11: $\mathbf{P}_{xz,k|k-1} = \mathbf{P}_{k|k-1} \mathbf{H}_{k|k-1}^T$
- 12: $\mathbf{P}_{z,k|k-1} = \mathbf{H}_{k|k-1} \mathbf{P}_{k|k-1} \mathbf{H}_{k|k-1}^T + \frac{1}{2} [\text{tr}(\mathbf{H}_{i,k}^* \mathbf{P}_{k|k-1} \mathbf{H}_{j,k}^* \mathbf{P}_{k|k-1})]_{ij}$
- 13: $\mathbf{S}_{k|k-1} = \mathbf{P}_{z,k|k-1} + \mathbf{R}_k$
- 14: $\mathbf{K}_k = \mathbf{P}_{xz,k|k-1} \mathbf{S}_{k|k-1}^{-1}$
- 15: $\hat{\mathbf{x}}_{k|k} = \hat{\mathbf{x}}_{k|k-1} + \mathbf{K}_k (\mathbf{z}_k - \hat{\mathbf{z}}_{k|k-1})$
- Recalibrate:**
- 16: $\mathbf{H}_{k|k} = \frac{\partial \mathbf{h}}{\partial \mathbf{x}}|_{\hat{\mathbf{x}}_{k|k}}$
- 17: $\mathbf{H}_{i,k}^* = [\frac{\partial^2 \mathbf{h}_i}{\partial x_m \partial x_n}|_{\hat{\mathbf{x}}_{k|k}}]_{mn}, i = 1, \dots, n_m$
- 18: $\mathbf{P}_{xz,k|k} = \mathbf{P}_{k|k-1} \mathbf{H}_{k|k}^T$
- 19: $\mathbf{P}_{z,k|k} = \mathbf{H}_{k|k} \mathbf{P}_{k|k-1} \mathbf{H}_{k|k}^T + \frac{1}{2} [\text{tr}(\mathbf{H}_{i,k}^* \mathbf{P}_{k|k-1} \mathbf{H}_{j,k}^* \mathbf{P}_{k|k-1})]_{ij}$
- 20: $\mathbf{S}_{k|k} = \mathbf{P}_{z,k|k} + \mathbf{R}_k$
- 21: $\mathbf{P}_{k|k} = \mathbf{P}_{k|k-1} + \mathbf{K}_k \mathbf{S}_{k|k} \mathbf{K}_k^T - \mathbf{P}_{xz,k|k} \mathbf{K}_k^T - \mathbf{K}_k \mathbf{P}_{xz,k|k}^T$
- Back out:**
- 22: **if** $\text{tr}(\mathbf{P}_{k|k}) > \text{tr}(\mathbf{P}_{k|k-1})$ **then**
- 23: $\hat{\mathbf{x}}_{k|k} = \hat{\mathbf{x}}_{k|k-1}$
- 24: $\mathbf{P}_{k|k} = \mathbf{P}_{k|k-1}$
- 25: **end if**
- 26: **end for**

Algorithm 3 The cubature Kalman filter with the new framework

Input: Process noise covariance matrix $\mathbf{Q}_k$, Measurement noise covariance matrix $\mathbf{R}_k$, states number n_x , state transition function $\mathbf{f}(\mathbf{x}, \mathbf{u})$, measurement function $\mathbf{h}(\mathbf{x})$, system inputs $\mathbf{u}_k$, measurements $\mathbf{z}_k$

Initialization:

- 1: $\hat{\mathbf{x}}_{0|0} = \mathbb{E}[\mathbf{x}_0]$
- 2: $\mathbf{P}_{0|0} = \mathbb{E}[(\hat{\mathbf{x}}_{0|0} - \mathbf{x}_0)(\hat{\mathbf{x}}_{0|0} - \mathbf{x}_0)^T]$
- 3: **for** every time step k **do**
- Predict:**
- 4: $\mathbf{L} = [\mathbf{L}_1, \dots, \mathbf{L}_{n_x}] = \text{chol}(\mathbf{P}_{k-1|k-1})$,
where $\mathbf{L}\mathbf{L}^T = \mathbf{P}_{k-1|k-1}$
- 5: $\mathcal{X}_{i,k-1} = \hat{\mathbf{x}}_{k-1|k-1} + \sqrt{n_x} \mathbf{L}_i, i = 1, \dots, n_x$
- 6: $\mathcal{X}_{i+n_x,k-1} = \hat{\mathbf{x}}_{k-1|k-1} - \sqrt{n_x} \mathbf{L}_i, i = 1, \dots, n_x$
- 7: $\mathcal{X}_{i,k|k-1}^* = \mathbf{f}(\mathcal{X}_{i,k-1}, \mathbf{u}_{k-1}), i = 1, \dots, 2n_x$
- 8: $\hat{\mathbf{x}}_{k|k-1} = \sum_{i=1}^{2n_x} \frac{1}{2n_x} \mathcal{X}_{i,k|k-1}^*$
- 9: $\mathbf{P}_{k|k-1} = \sum_{i=1}^{2n_x} \frac{1}{2n_x} [\mathcal{X}_{i,k|k-1}^* - \hat{\mathbf{x}}_{k|k-1}][\mathcal{X}_{i,k|k-1}^{*T} - \hat{\mathbf{x}}_{k|k-1}^T] + \mathbf{Q}_k$
- Update:**
- 10: $\mathbf{L} = [\mathbf{L}_1, \dots, \mathbf{L}_{n_x}] = \text{chol}(\mathbf{P}_{k|k-1})$
- 11: $\mathcal{X}_{i,k|k-1} = \hat{\mathbf{x}}_{k|k-1} + \sqrt{n_x} \mathbf{L}_i, i = 1, \dots, n_x$
- 12: $\mathcal{X}_{i+n_x,k|k-1} = \hat{\mathbf{x}}_{k|k-1} - \sqrt{n_x} \mathbf{L}_i, i = 1, \dots, n_x$
- 13: $\mathcal{Z}_{i,k|k-1} = \mathbf{h}(\mathcal{X}_{i,k|k-1}), i = 1, \dots, 2n_x$
- 14: $\hat{\mathbf{z}}_{k|k-1} = \sum_{i=1}^{2n_x} \frac{1}{2n_x} \mathcal{Z}_{i,k|k-1}$
- 15: $\mathbf{P}_{xz,k|k-1} = \sum_{i=1}^{2n_x} \frac{1}{2n_x} [\mathcal{X}_{i,k|k-1} - \hat{\mathbf{x}}_{k|k-1}][\mathcal{Z}_{i,k|k-1} - \hat{\mathbf{z}}_{k|k-1}]^T$
- 16: $\mathbf{P}_{z,k|k-1} = \sum_{i=1}^{2n_x} \frac{1}{2n_x} [\mathcal{Z}_{i,k|k-1} - \hat{\mathbf{z}}_{k|k-1}][\mathcal{Z}_{i,k|k-1} - \hat{\mathbf{z}}_{k|k-1}]^T$
- 17: $\mathbf{S}_{k|k-1} = \mathbf{P}_{z,k|k-1} + \mathbf{R}_k$
- 18: $\mathbf{K}_k = \mathbf{P}_{xz,k|k-1} \mathbf{S}_{k|k-1}^{-1}$
- 19: $\hat{\mathbf{x}}_{k|k} = \hat{\mathbf{x}}_{k|k-1} + \mathbf{K}_k(\mathbf{z}_k - \hat{\mathbf{z}}_{k|k-1})$
- Recalibrate:**
- 20: $\mathcal{X}_{i,k|k} = \hat{\mathbf{x}}_{k|k} + \sqrt{n_x} \mathbf{L}_i, i = 1, \dots, n_x$
- 21: $\mathcal{X}_{i+n_x,k|k} = \hat{\mathbf{x}}_{k|k} - \sqrt{n_x} \mathbf{L}_i, i = 1, \dots, n_x$
- 22: $\mathcal{Z}_{i,k|k} = \mathbf{h}(\mathcal{X}_{i,k|k}), i = 1, \dots, 2n_x$
- 23: $\hat{\mathbf{z}}_{k|k} = \sum_{i=1}^{2n_x} \frac{1}{2n_x} \mathcal{Z}_{i,k|k}$
- 24: $\mathbf{P}_{xz,k|k} = \sum_{i=1}^{2n_x} \frac{1}{2n_x} [\mathcal{X}_{i,k|k} - \hat{\mathbf{x}}_{k|k}][\mathcal{Z}_{i,k|k} - \hat{\mathbf{z}}_{k|k}]^T$
- 25: $\mathbf{P}_{z,k|k} = \sum_{i=1}^{2n_x} \frac{1}{2n_x} [\mathcal{Z}_{i,k|k} - \hat{\mathbf{z}}_{k|k}][\mathcal{Z}_{i,k|k} - \hat{\mathbf{z}}_{k|k}]^T$
- 26: $\mathbf{S}_{k|k} = \mathbf{P}_{z,k|k} + \mathbf{R}_k$
- 27: $\mathbf{P}_{k|k} = \mathbf{P}_{k|k-1} + \mathbf{K}_k \mathbf{S}_{k|k} \mathbf{K}_k^T - \mathbf{P}_{xz,k|k} \mathbf{K}_k^T - \mathbf{K}_k \mathbf{P}_{xz,k|k}^T$
- Back out:**
- 28: **if** $\text{tr}(\mathbf{P}_{k|k}) > \text{tr}(\mathbf{P}_{k|k-1})$ **then**
- 29: $\hat{\mathbf{x}}_{k|k} = \hat{\mathbf{x}}_{k|k-1}$
- 30: $\mathbf{P}_{k|k} = \mathbf{P}_{k|k-1}$
- 31: **end if**
- 32: **end for**

Algorithm 4 The unscented Kalman filter with the new framework

Input: Process noise covariance matrix $\mathbf{Q}_k$, Measurement noise covariance matrix $\mathbf{R}_k$, states number n_x , state transition function $\mathbf{f}(\mathbf{x}, \mathbf{u})$, measurement function $\mathbf{h}(\mathbf{x})$, system inputs $\mathbf{u}_k$, measurements $\mathbf{z}_k$, hyperparameters $\alpha = 10^{-3}, \beta = 2, \kappa = 0$

Initialization:

- 1: $\hat{\mathbf{x}}_{0|0} = \mathbb{E}[\mathbf{x}_0]$
- 2: $\mathbf{P}_{0|0} = \mathbb{E}[(\hat{\mathbf{x}}_{0|0} - \mathbf{x}_0)(\hat{\mathbf{x}}_{0|0} - \mathbf{x}_0)^T]$
- 3: $\lambda = \alpha^2(n_x + \kappa) - n_x$
- 4: $W_i^{(c)} = W_i^{(m)} = 1/(2n_x + 2\lambda), i = 1, \dots, n_x$
- 5: $W_0^{(c)} = \lambda/(n_x + \lambda) + 1 - \alpha^2 + \beta$
- 6: $W_0^{(m)} = \lambda/(n_x + \lambda)$
- 7: **for** every time step k **do**
- Predict:**
- 8: $\mathbf{L} = [\mathbf{L}_1, \dots, \mathbf{L}_{n_x}] = \text{chol}(\mathbf{P}_{k-1|k-1})$,
where $\mathbf{L}\mathbf{L}^T = \mathbf{P}_{k-1|k-1}$
- 9: $[\mathbf{L}_1, \dots, \mathbf{L}_{n_x}] = i^{\text{th}}$ column of $\mathbf{L}, i = 1, \dots, n_x$
- 10: $\mathcal{X}_{0,k-1} = \hat{\mathbf{x}}_{k-1|k-1}$
- 11: $\mathcal{X}_{i,k-1} = \hat{\mathbf{x}}_{k-1|k-1} + \sqrt{\lambda + n_x} \mathbf{L}_i, i = 1, \dots, n_x$
- 12: $\mathcal{X}_{i+n_x,k-1} = \hat{\mathbf{x}}_{k-1|k-1} - \sqrt{\lambda + n_x} \mathbf{L}_i, i = 1, \dots, n_x$
- 13: $\mathcal{X}_{i,k|k-1} = \mathbf{f}(\mathcal{X}_{i,k-1}, \mathbf{u}_{k-1}), i = 0, 1, \dots, 2n_x$
- 14: $\hat{\mathbf{x}}_{k|k-1} = \sum_{i=0}^{2n_x} W_i^{(m)} \mathcal{X}_{i,k|k-1}$
- 15: $\mathbf{P}_{k|k-1} = \sum_{i=0}^{2n_x} W_i^{(c)} [\mathcal{X}_{i,k|k-1} - \hat{\mathbf{x}}_{k|k-1}][\mathcal{X}_{i,k|k-1} - \hat{\mathbf{x}}_{k|k-1}]^T + \mathbf{Q}_k$
- Update:**
- 16: $\mathbf{L} = [\mathbf{L}_1, \dots, \mathbf{L}_{n_x}] = \text{chol}(\mathbf{P}_{k|k-1})$
- 17: $\mathcal{X}_{0,k|k-1} = \hat{\mathbf{x}}_{k|k-1}$
- 18: $\mathcal{X}_{i,k|k-1} = \hat{\mathbf{x}}_{k|k-1} + \sqrt{\lambda + n_x} \mathbf{L}_i, i = 1, \dots, n_x$
- 19: $\mathcal{X}_{i+n_x,k|k-1} = \hat{\mathbf{x}}_{k|k-1} - \sqrt{\lambda + n_x} \mathbf{L}_i, i = 1, \dots, n_x$
- 20: $\mathcal{Z}_{i,k|k-1} = \mathbf{h}(\mathcal{X}_{i,k|k-1}), i = 0, \dots, 2n_x$
- 21: $\hat{\mathbf{z}}_{k|k-1} = \sum_{i=0}^{2n_x} W_i^{(m)} \mathcal{Z}_{i,k|k-1}$
- 22: $\mathbf{P}_{xz,k|k-1} = \sum_{i=0}^{2n_x} W_i^{(c)} [\mathcal{X}_{i,k|k-1} - \hat{\mathbf{x}}_{k|k-1}][\mathcal{Z}_{i,k|k-1} - \hat{\mathbf{z}}_{k|k-1}]^T$
- 23: $\mathbf{P}_{z,k|k-1} = \sum_{i=0}^{2n_x} W_i^{(c)} [\mathcal{Z}_{i,k|k-1} - \hat{\mathbf{z}}_{k|k-1}][\mathcal{Z}_{i,k|k-1} - \hat{\mathbf{z}}_{k|k-1}]^T$
- 24: $\mathbf{S}_{k|k-1} = \mathbf{P}_{z,k|k-1} + \mathbf{R}_k$
- 25: $\mathbf{K}_k = \mathbf{P}_{xz,k|k-1} \mathbf{S}_{k|k-1}^{-1}$
- 26: $\hat{\mathbf{x}}_{k|k} = \hat{\mathbf{x}}_{k|k-1} + \mathbf{K}_k(\mathbf{z}_k - \hat{\mathbf{z}}_{k|k-1})$
- Recalibrate:**
- 27: $\mathcal{X}_{i,k|k} = \mathcal{X}_{i,k|k-1} + \mathbf{K}_k(\mathbf{z}_k - \hat{\mathbf{z}}_{k|k-1}), i = 0, \dots, 2n_x$
- 28: $\mathcal{Z}_{i,k|k} = \mathbf{h}(\mathcal{X}_{i,k|k}), i = 0, \dots, 2n_x$
- 29: $\hat{\mathbf{z}}_{k|k} = \sum_{i=0}^{2n_x} W_i^{(m)} \mathcal{Z}_{i,k|k}$
- 30: $\mathbf{P}_{xz,k|k} = \sum_{i=0}^{2n_x} W_i^{(c)} [\mathcal{X}_{i,k|k} - \hat{\mathbf{x}}_{k|k}][\mathcal{Z}_{i,k|k} - \hat{\mathbf{z}}_{k|k}]^T$
- 31: $\mathbf{P}_{z,k|k} = \sum_{i=0}^{2n_x} W_i^{(c)} [\mathcal{Z}_{i,k|k} - \hat{\mathbf{z}}_{k|k}][\mathcal{Z}_{i,k|k} - \hat{\mathbf{z}}_{k|k}]^T$
- 32: $\mathbf{S}_{k|k} = \mathbf{P}_{z,k|k} + \mathbf{R}_k$
- 33: $\mathbf{P}_{k|k} = \mathbf{P}_{k|k-1} + \mathbf{K}_k \mathbf{S}_{k|k} \mathbf{K}_k^T - \mathbf{P}_{xz,k|k} \mathbf{K}_k^T - \mathbf{K}_k \mathbf{P}_{xz,k|k}^T$
- Back out:**
- 34: **if** $\text{tr}(\mathbf{P}_{k|k}) > \text{tr}(\mathbf{P}_{k|k-1})$ **then**
- 35: $\hat{\mathbf{x}}_{k|k} = \hat{\mathbf{x}}_{k|k-1}$
- 36: $\mathbf{P}_{k|k} = \mathbf{P}_{k|k-1}$
- 37: **end if**
- 38: **end for**

Algorithm 5 The iterated extended Kalman filter with the old framework

Input: Process noise covariance matrix $\mathbf{Q}_k$, Measurement noise covariance matrix $\mathbf{R}_k$, state transition function $\mathbf{f}(\mathbf{x}, \mathbf{u})$, measurement function $\mathbf{h}(\mathbf{x})$, system inputs $\mathbf{u}_k$, measurements $\mathbf{z}_k$

Initialization:

- 1: $\hat{\mathbf{x}}_{0|0} = \mathbb{E}[\mathbf{x}_0]$
- 2: $\mathbf{P}_{0|0} = \mathbb{E}[(\hat{\mathbf{x}}_{0|0} - \mathbf{x}_0)(\hat{\mathbf{x}}_{0|0} - \mathbf{x}_0)^T]$
- 3: **for** every time step k **do**
- Predict:**
- 4: $\mathbf{F}_k = \frac{\partial \mathbf{f}}{\partial \mathbf{x}}|_{\hat{\mathbf{x}}_{k-1|k-1}, \mathbf{u}_{k-1}}$
- 5: $\hat{\mathbf{x}}_{k|k-1} = \mathbf{f}(\hat{\mathbf{x}}_{k-1|k-1}, \mathbf{u}_{k-1})$
- 6: $\mathbf{P}_{k|k-1} = \mathbf{F}_k \mathbf{P}_{k-1|k-1} \mathbf{F}_k^T + \mathbf{Q}_k$
- Update:**
- 7: $\mathbf{x}_0 = \hat{\mathbf{x}}_{k|k-1}$
- 8: **for** $i = 1$ to 1000 **do**
- 9: $\mathbf{H}_{k|k-1} = \frac{\partial \mathbf{h}}{\partial \mathbf{x}}|_{\mathbf{x}_{i-1}}$
- 10: $\hat{\mathbf{z}}_{k|k-1} = \mathbf{h}(\mathbf{x}_{i-1}) + \mathbf{H}_{k|k-1}(\mathbf{x}_{i-1} - \hat{\mathbf{x}}_{k|k-1})$
- 11: $\mathbf{P}_{xz,k|k-1} = \mathbf{P}_{k|k-1} \mathbf{H}_{k|k-1}^T$
- 12: $\mathbf{P}_{z,k|k-1} = \mathbf{H}_{k|k-1} \mathbf{P}_{k|k-1} \mathbf{H}_{k|k-1}^T$
- 13: $\mathbf{S}_{k|k-1,i} = \mathbf{P}_{z,k|k-1} + \mathbf{R}_k$
- 14: $\mathbf{K}_{k,i} = \mathbf{P}_{xz,k|k-1} \mathbf{S}_{k|k-1,i}^{-1}$
- 15: $\mathbf{x}_i = \hat{\mathbf{x}}_{k|k-1} + \mathbf{K}_{k,i}(\mathbf{z}_k - \hat{\mathbf{z}}_{k|k-1})$
- 16: **if** $i > 1$ and $\|\mathbf{x}_i - \mathbf{x}_{i-1}\| > \|\mathbf{x}_{i-1} - \mathbf{x}_{i-2}\|$ **then**
- 17: $\mathbf{S}_{k|k-1,i} = \mathbf{S}_{k|k-1,i-1}$
- 18: $\mathbf{K}_{k,i} = \mathbf{K}_{k,i-1}$
- 19: $\mathbf{x}_i = \mathbf{x}_{i-1}$
- 20: **break**
- 21: **end if**
- 22: **if** $\max_j |1 - \mathbf{x}_i(j)/\mathbf{x}_{i-1}(j)| < 0.001$ **then**
- 23: **break**
- 24: **end if**
- 25: **end for**
- 26: $\mathbf{S}_{k|k-1} = \mathbf{S}_{k|k-1,i}$
- 27: $\mathbf{K}_k = \mathbf{K}_i$
- 28: $\hat{\mathbf{x}}_{k|k} = \mathbf{x}_i$
- 29: $\mathbf{P}_{k|k} = \mathbf{P}_{k|k-1} - \mathbf{K}_k \mathbf{S}_{k|k-1} \mathbf{K}_k^T$
- 30: **end for**

Note that rows 16–21 are added to prevent the state estimations from diverging. Namely, when the sign of divergence is detected (row 16), the newest updates of the states, Kalman gain, and measurement residual covariance matrix are withdrawn (rows 17–19). If such a withdrawal is not made, we find in the simulation that the estimation accuracy can be much lower. This characteristic also suggests that the algorithm is not very robust and stable.

D Supplementary results related to the “back out” step

Fig. 8, 9, D.1–D.3 compare the state estimation accuracy with and without the “back out” step. Additionally, Fig. 10, D.4–D.7 quantify how often the proposed “back out” step is activated in practice by reporting the percentage of state updates that trigger this safeguard.

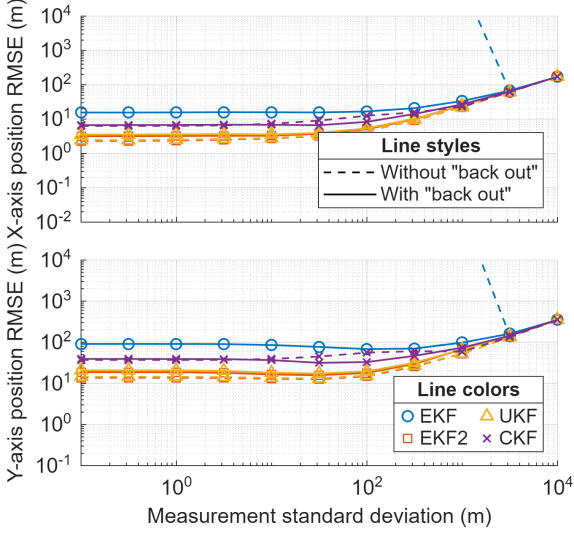


Fig. D.1. The root mean squared error of the state estimations with and without the “back out” step under different measurement noise setups (terrain-referenced navigation).

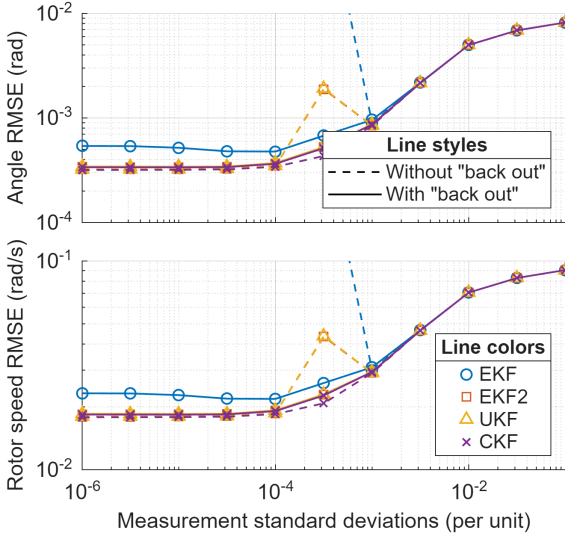


Fig. D.2. The root mean squared error of the state estimations with and without the “back out” step under different measurement noise setups (synchronous generator state estimation).

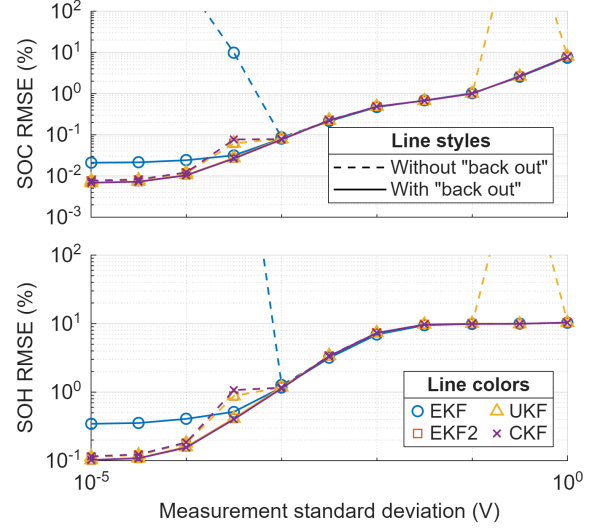


Fig. D.3. The root mean squared error of the state estimations with and without the “back out” step under different measurement noise setups (battery state estimation).

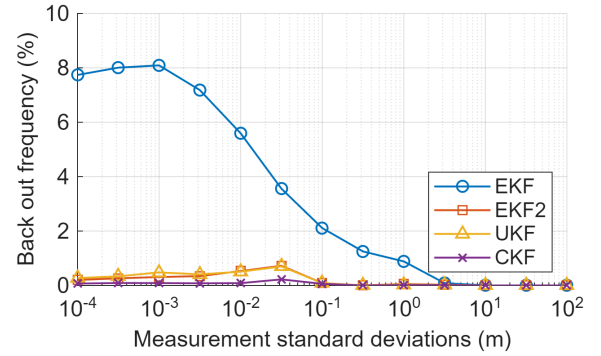


Fig. D.4. Percentage of state updates that trigger “back out” (3D target tracking).

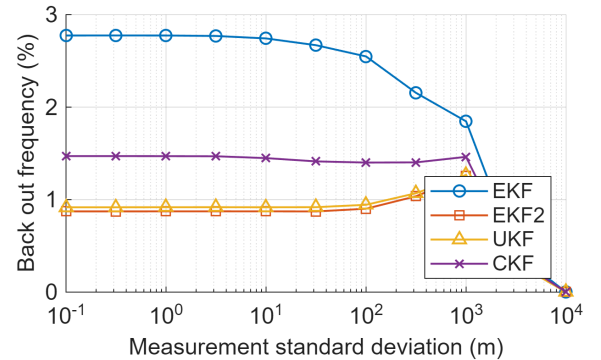


Fig. D.5. Percentage of state updates that trigger “back out” (terrain-referenced navigation).

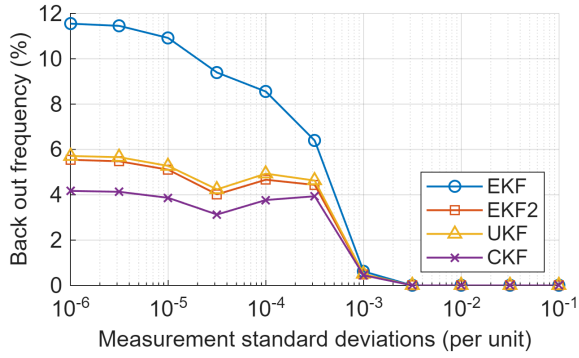


Fig. D.6. Percentage of state updates that trigger “back out” (synchronous generator state estimation).

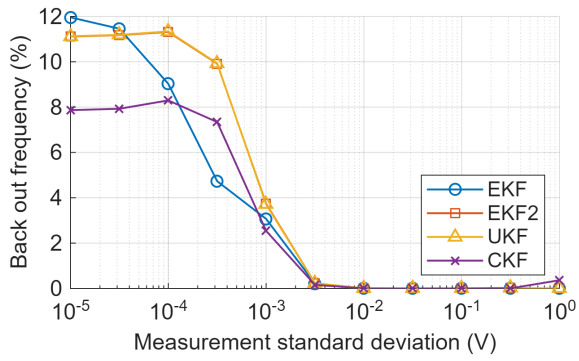


Fig. D.7. Percentage of state updates that trigger “back out” (battery state estimation).